Box-to-mask module

input: frozen features $\widetilde{F}$ and training boxes only

1 Whiten



PCA-whiten to 128 dims, project onto unit sphere

2 Background mixture



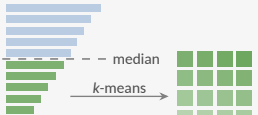
in-box
ring
clear

k-means

$K_{bg} = 12$ centroids G_j

3 Initialise foreground

in-box cells ranked by background likelihood



$K_{fg} = 16$ prototypes C_k

4 Expectation-Maximisation ×30

E-step per box

- **a** Subtract background log-likelihood from foreground log-likelihood $\rightarrow A$
- **b** Subtract within-box mean $\rightarrow \bar{A}$
- **c** Add spatial prior (log-odds) and apply a sigmoid $\rightarrow p_{fg}$

M-step all boxes

- Responsibility-weighted mean of cells per prototype
- Subtract matched outside-box centroid
- Re-estimate spatial prior as per-bin mean responsibility

responsibilities r_{nk}

prototypes C_k, spatial prior ρ

Fitted masker

1. whitening
2. background mixture
4. foreground prototypes
4. spatial prior



spatial prior (1 of 3 size bands)